

BnB'26 X DSC

Problem Statements

Disaster Management

[PSN001] Bio-Acoustic Swarm Sensing for Trapped Person Detection: Instead of a single radar unit, deploy a swarm of small, low-cost acoustic/vibration sensors that can be scattered or thrown across a collapse site. Each sensor picks up faint sounds (tapping, breathing, movement) and vibration signatures, then triangulates the likely location of survivors by cross-referencing signals across the swarm in real time, giving rescue teams a wider-coverage, faster-to-deploy alternative to a single directional radar device, especially useful when the collapse site is large or radar line-of-sight is blocked by dense debris.

[PSN002] AI-Powered Structural Stability Assessment: Create a mobile/simulation tool that helps first responders judge whether a partially collapsed building is safe to enter, estimating load-bearing capacity, flagging hazard zones, and mapping risk in real time using GIS-tagged photos and video.

[PSN003] Early Warning & Smart Resource Allocation System: Build a system combining live mapping and sensor/weather data to predict disaster impact zones and help authorities allocate ambulances, shelters, and relief supplies efficiently during a crisis.

Healthcare & Wellness

[PSN004] Early Screening Tool for Neurological Conditions: Build a low-cost, non-invasive screening aid usable by primary healthcare workers to flag early signs of conditions like Parkinson's or Alzheimer's and generate a referral recommendation for specialist care.

[PSN005] AI-Guided Digital Mental Health Platform: Create a system combining AI-guided conversational support, appointment booking, a peer-support community, and an admin dashboard so institutions can spot mental health trends and intervene early.

[PSN006] Offline EHR Companion for Rural Health Workers: Develop a mobile app letting community health workers manage patient records without internet access, auto-syncing when connectivity returns, with voice input support in local languages.

FinTech & Financial Inclusion

[PSN007] AI Credit Scoring for Underserved Borrowers: Design a model blending repayment history with alternative income signals (consumption patterns, utility payments) to fairly assess credit risk for people without formal financial records, enabling faster loan decisions.

[PSN008] Cybersecurity Framework for Rural Digital Banking: Build a lightweight fraud-detection and authentication system designed to run reliably on low-end smartphones and patchy networks, protecting rural users from digital banking fraud.

[PSN009] Micro-Savings Platform for Gig & Informal Workers: Create a tool that helps gig-economy and informal-sector workers automatically save small amounts based on irregular income patterns, with simple goal-tracking and withdrawal flexibility.

AI/ML & Emerging Tech

[PSN010] Multimodal Vision Extension for Open-Source LLMs: Enhance an open-source language model with image-understanding capability, enabling it to interpret pictures alongside text, with potential application to satellite/earth-observation data analysis.

[PSN011] AI-Powered Personalized Crop Advisory Assistant: Build an assistant combining weather, soil, and market-price data to give farmers real-time, personalized farming decisions through a simple conversational mobile interface.

[PSN012] AI-Driven Deepfake & Synthetic Media Detector: Develop a lightweight model that flags manipulated images, video, or audio in real time, helping platforms and users identify synthetic or altered media before it spreads.

Cybersecurity

[PSN013] Real-Time AI Phishing Detection System: Build a system using NLP and graph-based analysis to catch sophisticated phishing emails/links as they arrive (even ones designed to evade traditional filters) and block or flag them instantly.

[PSN014] Transformer-Based Web Application Firewall (WAF): Develop a pipeline that ingests live web-traffic logs, trains a transformer model to spot anomalies, and performs real-time detection of malicious requests, supporting both streaming and batch data.

[PSN015] Structured Vulnerability Assessment Challenge: Perform a hands-on security assessment of a given application: identify vulnerabilities, build proof-of-concept exploits, score each using CVSS, and recommend concrete remediation steps.

Gaming

[PSN016] Gamified Environmental Education Platform: Build an interactive, game-based app teaching students eco-friendly habits through challenges, rewards, and real-world tasks rather than textbook theory.

[PSN017] VR Training Simulator for High-Risk Skills: Use game-engine technology to build an immersive VR simulator (disaster response, industrial safety, or medical training) where users practice real procedures in a safe, virtual environment.

[PSN018] AI-Powered Adaptive Learning Game: Design a game that adjusts its difficulty and content in real time based on a player's performance, turning a school subject (math, coding, science) into a personalized, engaging learning loop.

Hardware

[PSN019] Smart Blind Stick for Visually Impaired Persons: Build an affordable, sensor-equipped walking stick that detects obstacles in real time, integrates GPS for navigation, and includes an emergency SOS button to improve mobility and independence.

[PSN020] Disaster Response Drone for Remote Areas: Design a drone system that delivers medical supplies and communication devices to disaster-hit or hard-to-reach areas, using AI for navigation and obstacle avoidance, paired with a coordination app.

[PSN021] Integrated Wearable for Rescue Worker Vital Monitoring: Build a wearable device tracking a rescue worker's vital signs, gas exposure, and fatigue in real time, wirelessly transmitting alerts to keep responders safe during hazardous operations.

Open Innovation

[PSN022] Open Innovation For Disaster Management: An open-ended brief for teams to identify their own disaster-preparedness, response, or recovery challenge and build a novel tech solution around it; no fixed problem given, judged purely on originality and impact.

[PSN023] Open Innovation For Healthcare & Wellness: Teams pick any unaddressed gap in healthcare delivery, wellness tracking, or patient experience and pitch their own original solution, free of a predefined ministry brief.

[PSN024] Open Innovation For FinTech & Financial Inclusion: An open track for teams to identify a real barrier to financial access or literacy and design their own product, spanning payments, lending, insurance, or savings innovation.

[PSN025] Open Innovation For AI/ML & Emerging Tech: Teams are free to apply AI, blockchain, AR/VR, or any emerging technology to a real-world problem of their choosing, with no predefined ministry brief, judged on technical depth and originality.

[PSN026] Open Innovation For Cybersecurity: An open track for teams to identify a real-world security gap (in an app, network, or system of their choosing) and build their own detection, prevention, or defense solution.

[PSN027] Open Innovation For Gaming: Teams design and build an original game or gamified experience (educational, social-impact, or entertainment-focused) around a theme of their own choosing.

[PSN028] Open Innovation For Hardware: An open-ended hardware brief for teams to design and prototype a physical device solving any real-world problem they identify, evaluated on functionality, feasibility, and innovation.